Project: ClaimWise – Automated Claims Processing System for Health Insurance

1. Introduction

This document outlines the Low-Level Design (LLD) for **ClaimWise**, an automated claims processing system designed for health insurance providers. The system digitizes claim submission, automates eligibility checks, manages approvals, and provides analytics for fraud detection and operational efficiency.

Supports backend development using **Java (Spring Boot)** and **.NET (ASP.NET Core)**.

2. Module Overview

2.1 Policyholder Profile & Coverage Module
2.2 Claim Submission & Document Upload Module
2.3 Eligibility & Rule Engine Module
2.4 Approval Workflow & Payout Module
2.5 Fraud Detection & Claims Analytics Module

3. Architecture Overview

3.1 Architectural Style

- **Frontend**: Angular/React for policyholder and admin portals.
- **Backend**: REST APIs for claims processing and rule evaluation.
- **Database**: SQL Server/PostgreSQL for structured insurance data.

3.2 Component Interaction

- Frontend captures claims and documents.
- Backend evaluates eligibility and processes approvals.

4. Module-Wise Design

4.1 Policyholder Profile & Coverage Module

4.1.1 Features

- View/update policyholder details and coverage limits.
- Track active policies and dependents.

4.1.2 Entities

- **Policyholder**
  - PolicyholderID
  - Name
  - PolicyType
  - CoverageDetails
  - Dependents

4.2 Claim Submission & Document Upload Module

4.2.1 Features

- Submit claims with hospital bills and discharge summaries.
- Upload documents via drag-and-drop interface.

4.2.2 Entities

- **Claim**
  - ClaimID
  - PolicyholderID
  - HospitalName
  - TreatmentDetails
  - Documents
  - Status

4.3 Eligibility & Rule Engine Module

4.3.1 Features

- Evaluate claim eligibility based on policy terms.
- Rule engine for exclusions, waiting periods, and limits.

4.3.2 Entities

- **EligibilityCheck**
  - CheckID
  - ClaimID
  - RuleApplied
  - Result

4.4 Approval Workflow & Payout Module

4.4.1 Features

- Multi-level approval based on claim amount.

- Track payout status and bank transfers.

4.4.2 Entities

- **Payout**
    - PayoutID
    - ClaimID
    - Amount
    - ApprovalStatus
    - TransferDate

4.5 Fraud Detection & Claims Analytics Module

4.5.1 Features

- Detect anomalies in claim patterns.
- Generate reports for audit and compliance.

4.5.2 Entities

- **ClaimsReport**
    - ReportID
    - Type (Fraud/Volume/Turnaround)
    - GeneratedDate
    - Insights

5. Deployment Strategy

5.1 Local Deployment

- Developer machines with sample claims data.

5.2 Staging and Production Environments

- Cloud deployment with secure document storage.

6. Database Design

6.1 Tables and Relationships

- Policyholder → Claim → EligibilityCheck → Payout → ClaimsReport

7. User Interface Design

7.1 Wireframes

- Policyholder Portal: Submit claims, view status.
- Admin Dashboard: Approvals, fraud alerts.
- Analytics Console: Claims trends and audit logs.

8. Non-Functional Requirements

8.1 Performance

- Handle 5,000 concurrent claims submissions.

8.2 Usability

- Simple UI for policyholders and agents.
- Tooltips and document upload guidance.

8.3 Security

- Encrypted documents and secure access tokens.
- Compliance with IRDAI and HIPAA standards.

8.4 Scalability

- Support for multiple insurance products and regions.

9. Assumptions and Constraints

9.1 Assumptions

- Hospitals will provide digital discharge summaries.
- Policyholders will submit claims via web portal.

9.2 Constraints

- Initial rollout for individual health insurance policies only.